SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

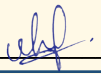
Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	**Vigneshwaran R**	**Reg. No.:**	**1911260109**
Date:		**Duration:**	60 minutes

Code of Conduct

I **Vigneshwaran R / 1911260109** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: 

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Annexure A – Solution: Agile Sprint Planning & User Story Workshop

Student: Vigneshwaran R **Reg. No.:** 1911260109 **Course:** CSA10 – Software Engineering

Problem Statement (Individual Project): *QuickBite – an Online Food Ordering & Delivery Application.* Customers in a city need one mobile/web platform to browse nearby restaurants, order food, pay online and track delivery, while restaurants manage menus and orders and delivery partners fulfil them. Today ordering is by phone, which causes wrong orders, long waits and no tracking.

1. Project Vision and Stakeholders

Vision: *For hungry city residents who want fast and reliable meals, QuickBite is a food ordering and delivery platform that lets them order from local restaurants in under 2 minutes and track the delivery live. Unlike phone ordering, QuickBite gives accurate orders, secure online payment and real-time status.*

Objectives (SMART):

- Allow a customer to complete an order (browse → cart → pay) in under 2 minutes.
- Onboard at least 50 restaurants and support 1,000 concurrent users within 6 months of launch.
- Achieve order accuracy of 98% or above and an average delivery time under 40 minutes.
- Keep payment success rate at 99% and app response time under 3 seconds.

Stakeholder	Role	Interest / Expectation
Customer	Primary user	Easy ordering, secure payment, live tracking, offers
Restaurant owner / staff	Primary user	Manage menu and prices, receive and accept orders quickly
Delivery partner	Primary user	Clear pickup/drop details, route help, fair earnings
Admin / Operations	Internal user	Approve restaurants, monitor orders, handle complaints
Product Owner	Scrum role	Owns backlog and priorities, maximises business value
Scrum Master	Scrum role	Facilitates ceremonies, removes impediments
Development Team (4 devs + 1 QA)	Scrum role	Design, build and test the increment
Payment gateway / Maps provider	External	Reliable APIs, security and compliance
Investors / Management	Sponsor	Revenue growth, market share, ROI

2. Epics and User Stories

Format: *As a <user>, I want <feature>, so that <benefit>.*

Epic	ID	User Story
E1 – User Account Management	US1	As a customer, I want to register and log in with my mobile number/email, so that I can place orders and save my details securely.
E2 – Restaurant Discovery	US2	As a customer, I want to browse nearby restaurants and view their menus, so that I can decide what to order.
	US3	As a customer, I want to search and filter by cuisine, price and rating, so that I can find food quickly.
E3 – Cart and Ordering	US4	As a customer, I want to add items to a cart and edit quantities, so that I can review my order before paying.

Epic	ID	User Story
	US5	As a customer, I want to place an order and pay online (UPI/card), so that my order is confirmed without cash handling.
E4 – Order Tracking and Notifications	US6	As a customer, I want to track my order status live on a map, so that I know when it will arrive.
	US10	As a customer, I want push notifications for each order stage, so that I stay informed without opening the app.
E5 – Restaurant and Delivery Operations	US7	As a restaurant owner, I want to add and update menu items and prices, so that customers always see correct information.
	US11	As a delivery partner, I want to accept delivery requests and see pickup/drop locations, so that I can deliver efficiently.
E6 – Reviews and Administration	US8	As a customer, I want to rate and review a delivered order, so that others can choose better restaurants.
	US9	As an admin, I want to approve and manage restaurants, so that only genuine restaurants appear on the platform.

3. Acceptance Criteria (Given–When–Then) for Sprint 1 Stories

US1 – Register / Login

- **Given** a new user on the sign-up screen, **when** they enter a valid mobile number and the correct OTP, **then** an account is created and they are logged in within 5 seconds.
- **Given** an existing user, **when** they enter wrong credentials 3 times, **then** the account is locked for 15 minutes and an error message is shown.
- **Given** a registered user, **when** they log in with valid details, **then** they land on the home screen with their saved address.

US2 – Browse Restaurants & Menu

- **Given** a logged-in user with location enabled, **when** the home screen opens, **then** restaurants within 10 km are listed with name, rating and delivery time in under 3 seconds.
- **Given** a list of restaurants, **when** the user taps one, **then** its menu with item name, price, image and veg/non-veg tag is displayed.
- **Given** an item is out of stock, **when** the menu loads, **then** it is shown as “Unavailable” and cannot be added.

US3 – Search & Filter

- **Given** the search bar, **when** the user types 3 or more characters, **then** matching restaurants/dishes appear within 2 seconds.
- **Given** search results, **when** the user applies cuisine, rating (≥4) and price filters, **then** only matching results remain and the result count is shown.

US4 – Cart

- **Given** a menu item, **when** the user taps “Add”, **then** the item is added and the cart badge and total update instantly.
- **Given** items in the cart, **when** the user changes quantity or removes an item, **then** the subtotal, taxes and total are recalculated correctly.
- **Given** items from restaurant A in the cart, **when** the user adds an item from restaurant B, **then** a prompt asks to clear the cart first.

US5 – Place Order & Online Payment

- **Given** a non-empty cart and a saved address, **when** the user pays via UPI/card successfully, **then** the order is confirmed, an order ID is generated and a confirmation is shown within 5 seconds.
- **Given** a failed or cancelled payment, **when** the gateway returns failure, **then** no order is created, the cart is retained and the user can retry.
- **Given** a confirmed order, **when** it is saved, **then** the restaurant receives it and card details are never stored in our database.

4. Product Backlog Prioritisation

Method: **MoSCoW** combined with **value vs. dependency** (business value 1–10, customer need, and whether other stories depend on it). Authentication, browsing, cart and payment are the core "order flow" that every other feature depends on, so they come first.

Rank	Story	Business value	Dependency	MoSCoW	Sprint
1	US1 Register / Login	9	None – needed by all	Must	1
2	US2 Browse restaurants & menu	10	US1	Must	1
3	US4 Cart	9	US2	Must	1
4	US5 Order & online payment	10	US4 (revenue-critical)	Must	1
5	US3 Search & filter	7	US2	Should	1
6	US7 Restaurant menu management	8	US1	Must	2
7	US11 Delivery partner accepts order	8	US5	Must	2
8	US6 Live order tracking	8	US5, US11	Should	2
9	US10 Push notifications	6	US5	Should	3
10	US9 Admin restaurant management	6	US1	Should	3
11	US8 Ratings & reviews	5	US5 (delivered order)	Could	3

5. Sprint 1 Backlog and Task Breakdown

Sprint length: 2 weeks (10 working days). Team: 4 developers + 1 QA. Estimated velocity: ~25 story points. Selected: US1, US2, US3, US4, US5 (24 points).

Story	Tasks	Est. (hrs)
US1 Register / Login	Design login/OTP screens (4); build OTP and auth API (8); JWT session handling (4); validation and lockout logic (4); unit + API tests (4)	24
US2 Browse & menu	Design restaurant list and menu UI (6); restaurant/menu DB schema (4); location-based listing API (8); menu API + stock flag (6); integrate UI with API (8); testing (4)	36
US3 Search & filter	Search API with indexing (8); filter logic – cuisine/rating/price (6); search UI with filters (6); testing (4)	24
US4 Cart	Cart data model (3); add/update/remove APIs (6); totals and tax calculation (4); cart UI and badge (6); single-restaurant rule + tests (4)	23
US5 Order & payment	Order data model (4); payment gateway integration – sandbox (12); order placement API (8); payment failure/retry flow (6); checkout and confirmation UI (8); security review – no card storage (4); end-to-end testing (8)	50

Total planned effort ≈ 157 hrs against team capacity of ≈ 160 hrs (4 devs × ~5 productive hrs/day × 10 days = 200 hrs, minus ~20% for ceremonies, reviews and QA support), so the sprint is realistic.

6. Story Point Estimation (Fibonacci / Planning Poker)

The team used **Planning Poker** with the Fibonacci scale (1, 2, 3, 5, 8, 13). **US1 (3 points)** was taken as the reference story; others were sized relative to it by complexity, effort and risk.

Story	Points	Justification
US1 Register / Login	3	Well-known pattern; OTP service is a ready library; low risk.
US2 Browse & menu	5	Needs DB design, location-based query and UI; moderate complexity.
US3 Search & filter	5	Indexing and multiple filter combinations; depends on US2 data.
US4 Cart	3	Simple CRUD plus total calculation; rules are clear.
US5 Order & payment	8	External gateway integration, failure handling and security; highest risk and uncertainty.
Total	**24**	Within team velocity of ~25 points.

7. Sprint Goal and Expected Deliverables

Sprint Goal: *"By the end of Sprint 1, a registered customer can log in, find a restaurant, add dishes to a cart and place a paid order, giving us a working end-to-end ordering flow."*

Expected deliverables (potentially shippable increment):

- Working login/registration with OTP verification.
- Restaurant listing, menu view, search and filters.
- Shopping cart with price, tax and total calculation.
- Checkout with online payment (sandbox) and order confirmation with order ID.
- Test report, updated product backlog and sprint burndown chart.

Definition of Done: code reviewed and merged, unit and integration tests passed, all acceptance criteria verified by QA, no critical bugs, deployed on the staging server and demonstrated to the Product Owner in the Sprint Review.

Ceremonies: Sprint Planning (Day 1), Daily Stand-up (15 min), Sprint Review and Retrospective (Day 10).